

ULTIMATE PROBABILITY & STATISTICS

Formula-first revision sheet - what to use, when to use it, and the traps that lose marks

1. Classify the data Ungrouped? Discrete frequency? Continuous class intervals? Probability table/tree?	2. Build the working table For stats: x , f , CF , midpoint. For probability: events, complements, branch probabilities.	3. Interpret last State units, direction of skewness, tail type, or what the conditional probability means.
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Notation you must define once

Use $N = \sum f$ for grouped data; x is a value or class midpoint.

Formula / rule	Use it when... / exam move
x , $\bar{x}$, f , N	x = observation or class midpoint; $\bar{x}$ = mean; f = frequency; $N = \sum f$.
L , h , $CF<$, f_m	For grouped interpolation: L = lower limit of target class, h = class width, $CF<$ = cumulative frequency before it, f_m = target-class frequency.
Q_1 , Q_2 , Q_3 ; D_k ; P_k	Quartiles, deciles, percentiles. Q_2 is the median.
A , d , u	Assumed mean A ; $d = x - A$; $u = d / h$ for step-deviation.
$P(A)$, $P(B A)$, $A \cap B$	Probability of A ; probability of B given A ; intersection = both events.

Fast formula selector

The wording often tells you the method before any calculation begins.

If the question says...	Go to...
Typical value, no serious outliers	Mean. Use weighted mean if observations have different importance.
Middle / typical value with outliers, skewed data	Median. For continuous grouping, use the interpolation formula.
Most common category/value	Mode. It is the only central measure usable for nominal categories.
Top $x\%$, bottom $x\%$, SLA covers 95%	Percentile: top $x\% = P(100-x)$; bottom $x\% = P_x$; SLA threshold often P_{95} or P_{99} .
Consistency / compare two systems on different scales	Coefficient of variation. Lower CV means more consistent.
Direction of asymmetry / tail risk	Skewness and kurtosis (usually after calculating central moments).
Given that / among those / conditional table	Conditional probability. Denominator is the total of the GIVEN condition.
Causes form cases; need overall chance	Law of total probability. Multiply down each case, then add.
Observed effect; need likely cause	Bayes theorem. Numerator = one cause's branch; denominator = all matching branches.

1. Central tendency

Before grouped-data work, calculate $N = \sum f$. For continuous classes, x means midpoint.

Formula / rule	Use it when... / exam move
$\bar{x} = \sum x / n$	Ungrouped arithmetic mean. Use for a typical value when all observations count equally.
$\bar{x} = \sum (fx) / \sum f$	Discrete frequency or continuous grouped mean. For continuous intervals, first take midpoint $x = (\text{lower} + \text{upper}) / 2$.
$\bar{x}_w = \sum (wx) / \sum w$	Weighted mean. Use when traffic, credits, importance, or sample sizes differ. Never divide by number of groups unless weights are equal.
$\bar{x} = A + \sum d / n$	Assumed-mean shortcut, ungrouped. Take $d = x - A$. Choose a convenient central A .
$\bar{x} = A + \sum (fd) / \sum f$	Assumed-mean shortcut, grouped. Same result as direct method; reduces arithmetic.
$\bar{x} = A + [\sum (fu) / \sum f] h$	Step deviation. Use when midpoint differences share class width h ; $u = (x-A)/h$.
correct mean = wrong mean + $\sum (\text{correct} - \text{wrong}) / n$	Correcting a mean. Convert the original mean to its total first, then replace every misread value. Works for one or many errors.
Median, odd n : value at $(n+1)/2$	Ungrouped data. Sort first. Position, not original order, determines the median.
Median, even n : $[x(n/2) + x(n/2+1)] / 2$	Ungrouped even n. Sort first, then average the two middle observations.
Discrete grouped median: value whose CF first reaches $(N+1)/2$	Discrete frequency table. Make CF, find the position, and report the corresponding x value - no interpolation.
Median = $L + [(N/2 - CF_{<}) / f_m] h$	Continuous grouped median. Find the class whose CF first reaches $N/2$, then interpolate. It is an estimate.
Mode = value with largest f	Ungrouped or discrete grouped. Can be none or more than one. Use for most frequent category/value.
Mode = $L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] h$	Continuous grouped mode. f_1 = modal-class frequency; f_0 and f_2 are preceding/following class frequencies.

2. Positional averages

Locate the target position with CF, then interpolate only for continuous grouped data.

Formula / rule	Use it when... / exam move
Ungrouped Q_k position = $k(n+1)/4$	$k = 1, 2, 3$. Sort the data. If position is fractional, linearly interpolate between adjacent values.
Ungrouped D_k position = $k(n+1)/10$	$k = 1$ to 9 . D_5 is the median (50th percentile).
Ungrouped P_k position = $k(n+1)/100$	$k = 1$ to 99 . The course notes use this positional convention; follow the exam's stated convention if it differs.
Grouped $Q_k = L + [(kN/4 - CF_{<}) / f_Q] h$	Find the class whose CF reaches $kN/4$. $Q_1 = 25\text{th}$ percentile, $Q_2 = \text{median}$, $Q_3 = 75\text{th}$ percentile.
Grouped $D_k = L + [(kN/10 - CF_{<}) / f_D] h$	Find the class whose CF reaches $kN/10$.
Grouped $P_k = L + [(kN/100 - CF_{<}) / f_P] h$	Find the class whose CF reaches $kN/100$. P_{95}/P_{99} are common SLA cutoffs.
$IQR = Q_3 - Q_1$; $QD = (Q_3 - Q_1)/2$	Middle spread. Use IQR/QD when outliers make range or SD misleading. Middle 50% lies from Q_1 to Q_3 .

3. Dispersion and consistency

Use an absolute measure for actual spread; use a coefficient to compare different scales.

Formula / rule	Use it when... / exam move
Range = maximum - minimum	Quick rough spread. Highly sensitive to outliers; not suitable for open-ended classes.
Coefficient of range = $(\text{max}-\text{min})/(\text{max}+\text{min})$	Unit-free version of range. Use to compare relative ranges across scales.
MD about mean = $\sum x - \bar{x} / n$	Ungrouped mean deviation. Use absolute values because signed deviations from the mean sum to 0.
MD grouped = $\sum [f x - \bar{x}] / \sum f$	Grouped mean deviation. In continuous data use midpoint x .
Coefficient of MD = MD / mean	Use if mean deviation has already been calculated and a relative comparison is requested.
Population variance: $\sigma^2 = \sum (x - \mu)^2 / N$	Use for a complete population. Units are squared.
Sample variance: $s^2 = \sum (x - \bar{x})^2 / (n-1)$	Use for a sample when the question explicitly asks for sample variance/SD.
Grouped variance = $\sum [f(x - \bar{x})^2] / \sum f$	Course population-style grouped variance. Use midpoint x for continuous classes.
$\sigma = \sqrt{\text{variance}}$	Standard deviation: preferred spread measure; same units as data and gives extra weight to large deviations.
variance = $\sum (x^2)/n - (\bar{x})^2$	Computational shortcut, ungrouped. Useful when $\sum x$ and $\sum x^2$ are given. For grouped data use $\sum (fx^2)/N - (\bar{x})^2$.
CV = $(\text{SD} / \text{mean}) \times 100\%$	Compare consistency. Lower CV = less relative variability = more consistent. Do not choose using SD alone when means/scales differ.
Coefficient of QD = $(Q3 - Q1)/(Q3 + Q1)$	Relative, robust spread. Use when data are skewed or contain outliers.

4. Skewness, kurtosis, and central moments

For a moments question, make columns for d , d^2 , d^3 , d^4 (and multiply each by f when grouped).

Formula / rule	Use it when... / exam move
Karl Pearson $Sk = (\text{mean} - \text{mode})/\text{SD}$	Use when a clear unique mode exists. $Sk > 0$ right skew; $Sk < 0$ left skew; $Sk = 0$ symmetric.
Karl Pearson $Sk = 3(\text{mean} - \text{median})/\text{SD}$	Use when mode is missing, unreliable, or multimodal. This is often the safer exam formula.
Bowley $Sk = (Q3 + Q1 - 2Q2)/(Q3 - Q1)$	Quartile-based and robust to outliers. Between -1 and +1. Use when $Q1$, median, $Q3$ are given.
$\mu_r = \sum [(x - \bar{x})^r] / n$	Central moment, ungrouped. For grouped data: $\mu_r = \sum [f(x - \bar{x})^r] / \sum f$.
$\mu_1 = 0$; $\mu_2 = \text{variance}$; $\text{SD} = \sqrt{\mu_2}$	First check: μ_1 must be zero (apart from rounding). Second moment measures spread.
$\gamma_1 = \mu_3 / \mu_2^{(3/2)}$	Moment skewness, signed. Positive = right skew; negative = left skew.
$\beta_1 = \mu_3^2 / \mu_2^3$	Moment coefficient of skewness. Non-negative; use γ_1 's sign from μ_3 when interpretation asks direction.
$\beta_2 = \mu_4 / \mu_2^2$	Kurtosis. $\beta_2 = 3$ mesokurtic; >3 leptokurtic (heavy tails); <3 platykurtic (light tails).
excess kurtosis = $\beta_2 - 3$	Use only if the question explicitly asks for excess kurtosis. Normal distribution then has excess 0.

Formula / rule	Use it when... / exam move
Mean = median = mode: symmetric	Mean > median > mode: positive/right skew. Mean < median < mode: negative/left skew.

5. Counting and basic probability

Translate first: AND across stages = multiply; OR across distinct cases = add; order matters = permutation.

Formula / rule	Use it when... / exam move
Total outcomes = $n_1 \times n_2 \times \dots \times n_k$	Fundamental counting principle. Use for sequential choices/stages. Apply restrictions by splitting into non-overlapping cases, then add.
$n! = n(n-1)\dots 1$; $0! = 1$	Foundation for permutations/combinations.
$nPr = n!/(n-r)!$	Permutation. Use when positions, awards, ranks, or sequences differ: order matters.
$nCr = n!/[r!(n-r)!]$	Combination. Use when selecting a team, hand, features, or subset: order does not matter.
Distinct arrangements = $n!/(n_1! n_2! \dots n_k!)$	Use when some objects are identical, e.g. letters in a word. Divide by factorial for each repeated group.
$P(A) = \text{favourable outcomes} / \text{total outcomes}$	Classical probability: only when all listed outcomes are equally likely.
$0 \leq P(A) \leq 1$; $P(S) = 1$	Sanity check for every answer. A negative probability or one above 1 is impossible.
$P(A') = 1 - P(A)$	Complement. Best for 'at least one', 'none', 'not', or a long OR list.
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	General addition/inclusion-exclusion. Use for A OR B when overlap is possible; subtract the overlap once.
If $A \cap B$ is empty: $P(A \cup B) = P(A) + P(B)$	Add directly only for mutually exclusive events. 'Independent' does NOT mean mutually exclusive.
$P(\text{at least one}) = 1 - P(\text{none})$	For independent repeated trials with success p : $P(\text{at least one success}) = 1 - (1-p)^n$.

6. Conditional probability and product rule

A condition after the vertical bar changes the denominator - restrict the sample space to that condition.

Formula / rule	Use it when... / exam move
$P(B A) = P(A \cap B) / P(A)$	Conditional probability. Use for 'given A', 'among A', a row/column of a contingency table. Requires $P(A) > 0$.
$P(A \cap B) = P(A) P(B A)$	General product rule. Use for A AND B, especially dependent sequential draws / tree branches.
$P(A \cap B) = P(B) P(A B)$	Same intersection, reverse order. Pick the conditional probability provided in the question.
Independent iff $P(A \cap B) = P(A)P(B)$	Equivalent to $P(B A) = P(B)$, provided $P(A) > 0$. With replacement is often independent; without replacement is usually dependent.
Without replacement: update counts	After a first draw, reduce total by 1 and reduce the selected type by 1 if it was drawn. Multiply branch probabilities.
Contingency table: $P(\text{row} \text{column}) = \text{cell} / \text{column total}$	The denominator is the total named after 'given'. For $P(\text{male} \text{secondary})$, divide by total secondary - not grand total.
$P(A \text{ only}) = P(A) - P(A \cap B)$	Use for 'A but not B'. For 'neither A nor B', use $1 - P(A \cup B)$.

7. Law of total probability and Bayes theorem

Draw a two-stage tree whenever there are sources/causes and a result. Multiply along branches; add matching endpoints.

Formula / rule	Use it when... / exam move
$P(A) = \sum P(B_i) P(A B_i)$	Law of total probability. Use when $B_1 \dots B_k$ are mutually exclusive and exhaustive causes (a partition), and you need overall $P(A)$.
$P(B_r A) = \frac{P(B_r)P(A B_r)}{\sum [P(B_i)P(A B_i)]}$	Bayes theorem. Use when evidence A is observed and the question asks which source/cause B_r is most likely.
Bayes numerator = prior x likelihood	For one candidate cause: $P(B_r) \times P(A B_r)$. Do not omit the prior/base rate.
Bayes denominator = $P(A)$	Get it by total probability: add every branch that ends in observed evidence A. Posterior probabilities across all causes must sum to 1.
Diagnostic test: $P(\text{disease} \text{positive})$	Use prevalence as prior, sensitivity as $P(\text{positive} \text{disease})$, and false-positive rate as $P(\text{positive} \text{no disease})$. Low prevalence can make a positive result much less certain than expected.
Sequential inspection / stage reached	To be rejected at stage 3, multiply: survive 1 x survive 2 x reject 3. Include all prior conditions needed to reach a stage.

8. Exam recipes and mark-saving checks

Write the structure, then substitute numbers. A correct formula with clearly identified symbols earns method marks.

Formula / rule	Use it when... / exam move
Grouped statistics	1) Make x (midpoint), f , fx , CF . 2) Check sum $f = N$. 3) Identify target class from CF . 4) Use L , $CF<$, f , h . 5) State result is an estimate for continuous grouping.
Moments question	1) Find mean. 2) Make $d = x - \bar{x}$. 3) compute $f d^2$, $f d^3$, $f d^4$. 4) Divide sums by N for μ_2 , μ_3 , μ_4 . 5) calculate SD, skewness, β_2 and interpret.
Correction of mean	Wrong total = $n \times$ wrong mean. Correct total = wrong total - wrong entries + true entries. Correct mean = correct total/ n .
Two-event probability	Write 'OR' as union and 'AND' as intersection. Use general addition unless the events really cannot happen together.
Conditional table	Circle/underline the word after 'given'. Sum that row/column for denominator; cell overlap is numerator.
Bayes/tree	Branches start with priors, then conditional probabilities. Multiply down. Add matching leaf branches for evidence. Divide desired branch by evidence total.
Common traps	Do not: use $N+1$ over 2 inside the continuous median formula; use mode formula for discrete data; compare SD instead of CV across scales; add overlapping probabilities; confuse $P(A B)$ with $P(B A)$; forget complementary branches sum to 1.
Final line of an answer	Give units and meaning: 'P95 = ... ms, so 95% of observations are at or below this value'; 'CV A < CV B, so A is more consistent'; 'Sk > 0, so right-skewed.'

Coverage: lectures 1-14 and worksheets 1-3. This sheet preserves the course conventions for grouped data while flagging the conditions needed before simplifying a probability rule.